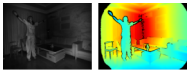
five sensing streams (one training clip: jumping jacks)

IR + depth video

640 x 480, 16 frames



infrared, depth

thermal video

320 x 240, 16 frames



heat of body and objects

skeleton

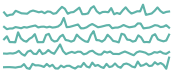
17 joints, 3-D



tracked joints, world frame

five inertial units

wrists, ankles, chest



|acceleration|, 5 devices

radar

session key; scene repeated 3x



YOLO11n crops

specialists (accuracy alone, OOF)

person crop, 384 px
MViTv2-S
21.6 MB — **0.733**

wrist crop, 224 px
MViTv2-S
21.6 MB — **0.720**

thermal frame, 224 px
MViTv2-S
21.6 MB — **0.634**

skeleton + IMU
20-member ensemble
18.9 MB — **0.618**

IMU statistics
ExtraTrees forest
1.6 MB — **0.377**

fused 0.781

prior-corrected log-linear fusion

$$\sum_i w_i \log p_i$$

coefficients by nested CV

decoded 0.830

ordered transition decoder

activity-script prior, label-free, per recording

recording group, clip order

5
sensors

40
activities

96 MB
one file

0.877
private score

unseen people
14th of 307 teams
0/405 changed
on 5 GPUs

accuracies: pooled out-of-fold over 2,700 training clips of held-out subjects. private: Kaggle leaderboard, 204 clips of unseen subjects.